

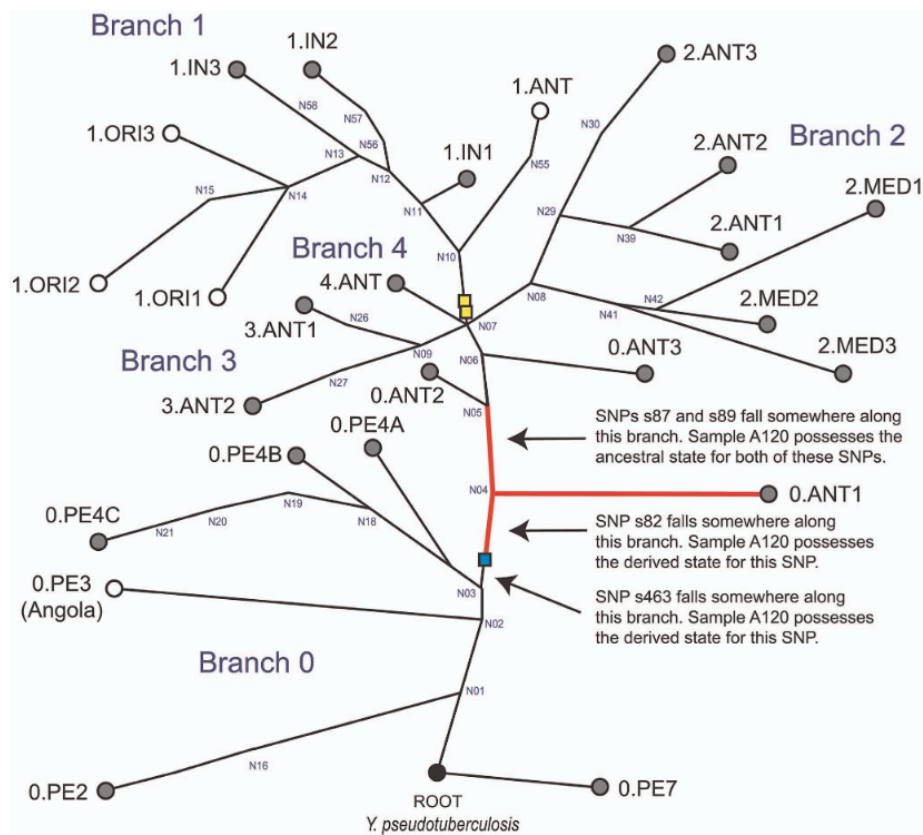
historical epidemiology of plague

1 December 2025

plague: first pandemic

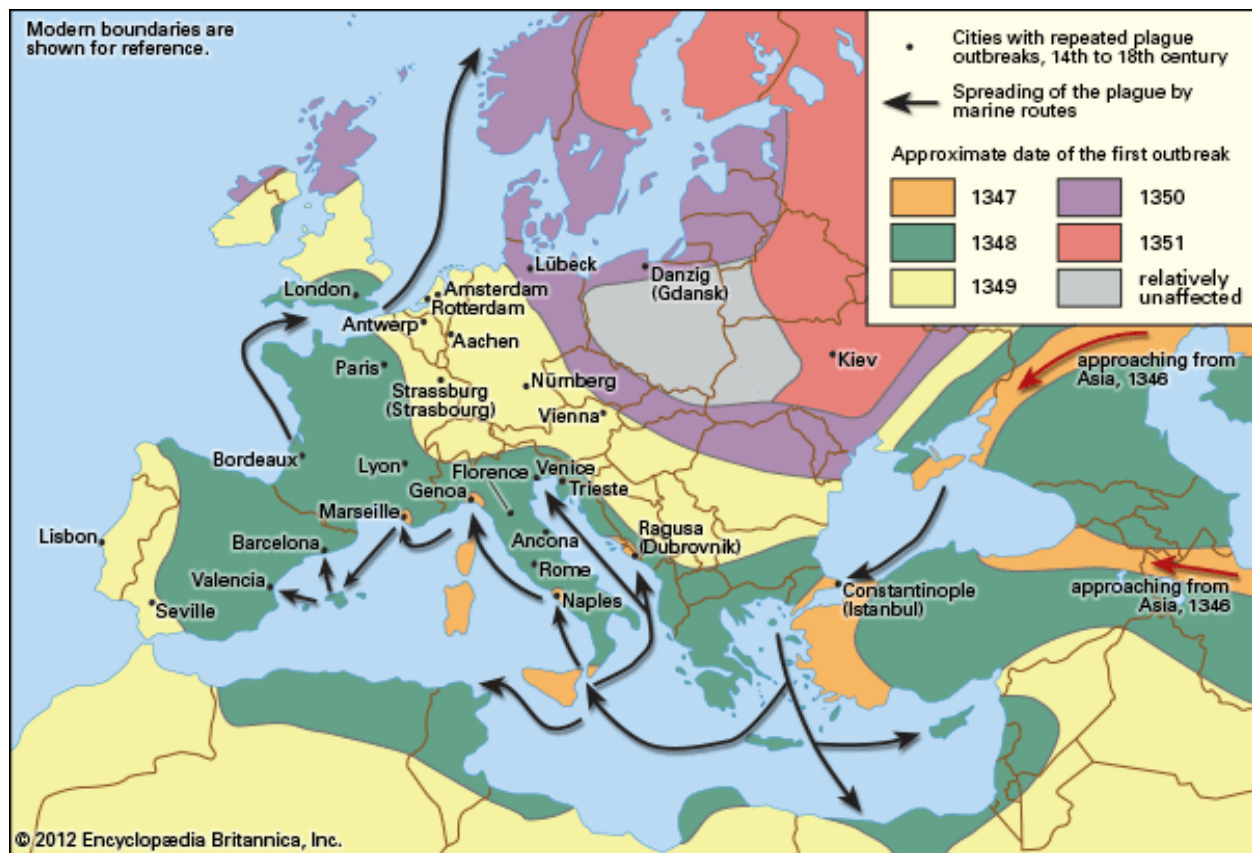
- *Yersinia pestis*
- causative agent of the *Black Death*
- Plague of Justinian (541) - approx. 750 (ancient DNA)

Harbeck et al. (2013)



second pandemic

- Black Death! 1347 onward
- human pop 450M → 350-375M by 1400
- continues until late 18th c.
- Shakespeare, Isaac Newton, etc. etc.
- also validated from ancient DNA



Britannica

acceleration (Earn et al., 2020)

- get historical details on plague from wills, parish records, London Bills of Mortality ...

The Diseases and Casualties this Week.

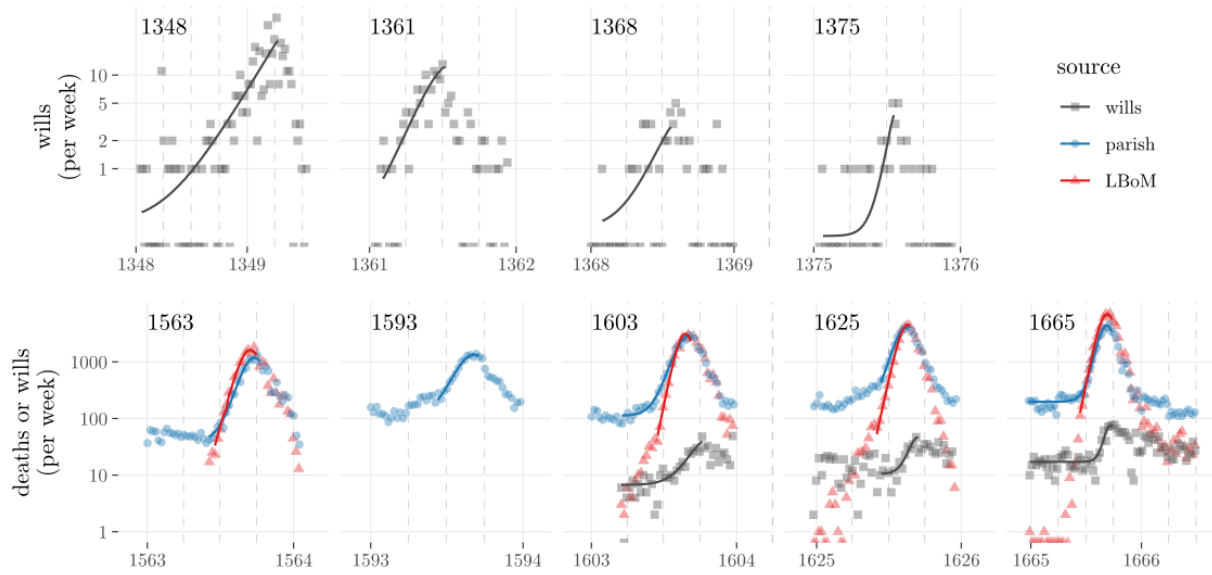
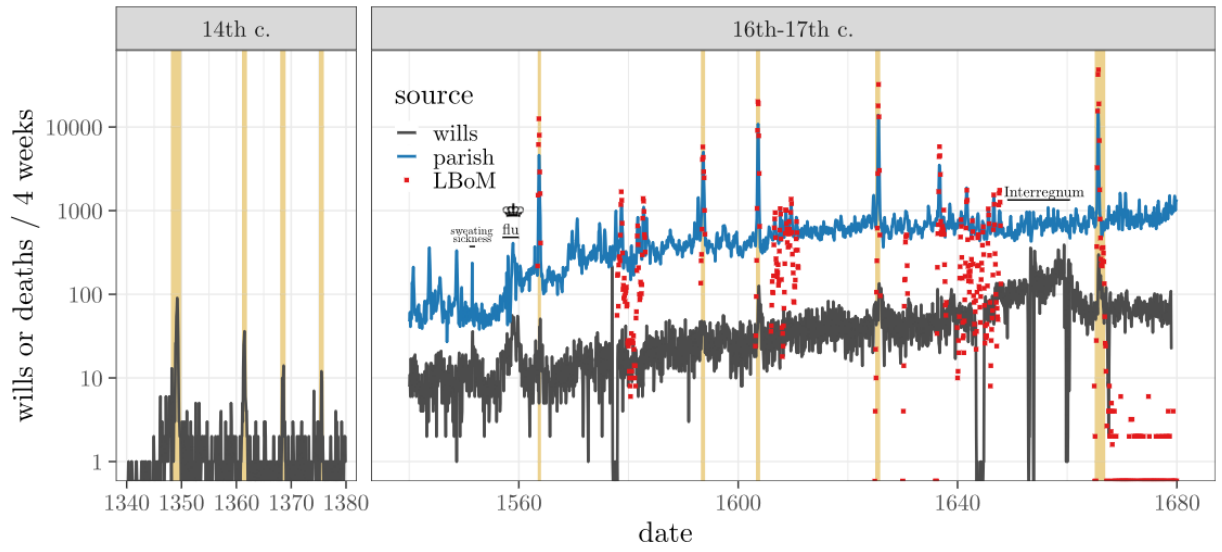


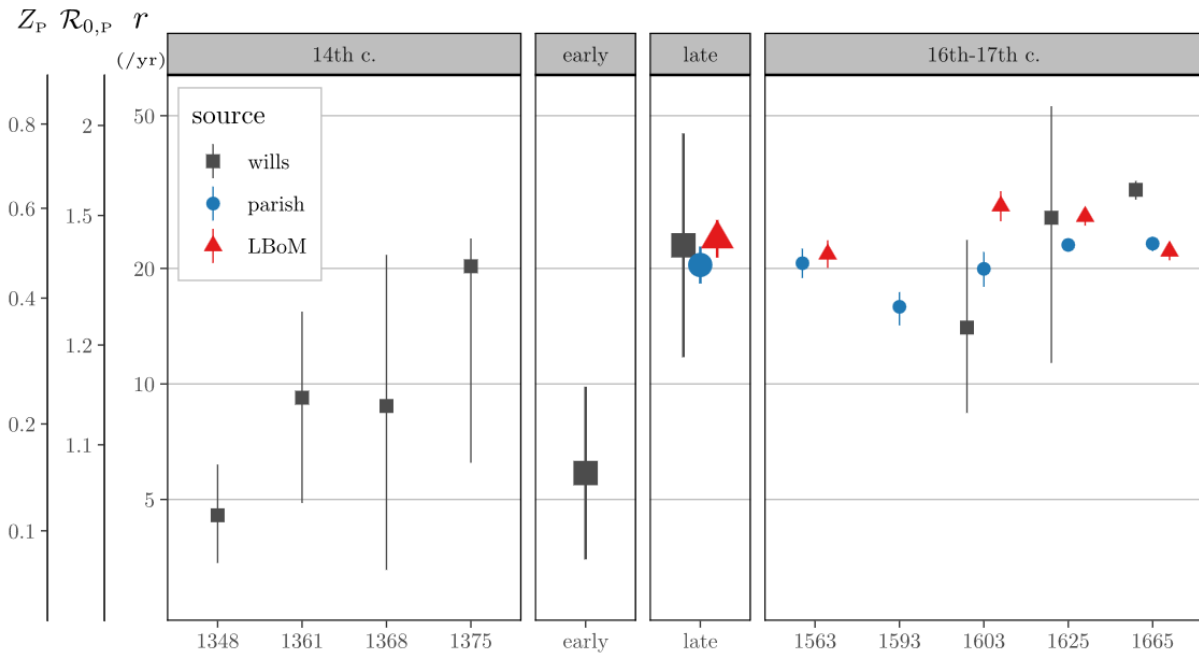
A Bortive	6	Frighted	1
Aged	50	Gout	1
Ague	1	Grief	3
Apoplexie	2	Griping in the Guts	35
Childbed	42	Jaundies	2
Chrisomes	11	Imposthume	8
Cold	1	Infants	9
Consumption	99	Kingevil	2
Convulsion	63	Meagrome	2
Cough	1	Plague	5533
Dropsie	22	Purples	2
Drownd at St. Martin in the Fields	1	Rickets	10
Feaver	268	Rising of the Lights	13
Fistula	2	Rupture	1
Flox and Small-pox	4	Scurvy	5
Flux	1	Spotted Feaver	65
Found dead in the Fields at St. Mary Islington	1	Stilborn	10
		Stone	3
		Stopping of the Stomach	6
		Suddenly	1
		Surfeit	36
		Teeth	112
		Thrush	3
		Tifick	5
		Vomiting	4
		Winde	1
		Wormes	12

Christned	Males — 68	Buried	Males — 3212	Plague — 5533
	Females — 78		Females — 3248	
	In all — 146		In all — 6460	

Decreased in the Burials this Week — 1837
 Parishes clear of the Plague — 7 Parishes Infected — 123

*The Assize of Bread set forth by Order of the Lord Mayor and Court of Aldermen,
 A penny Wheaten Loaf to contain Nine Ounces and a half, and three
 half-penny White Loaves the like weight.*



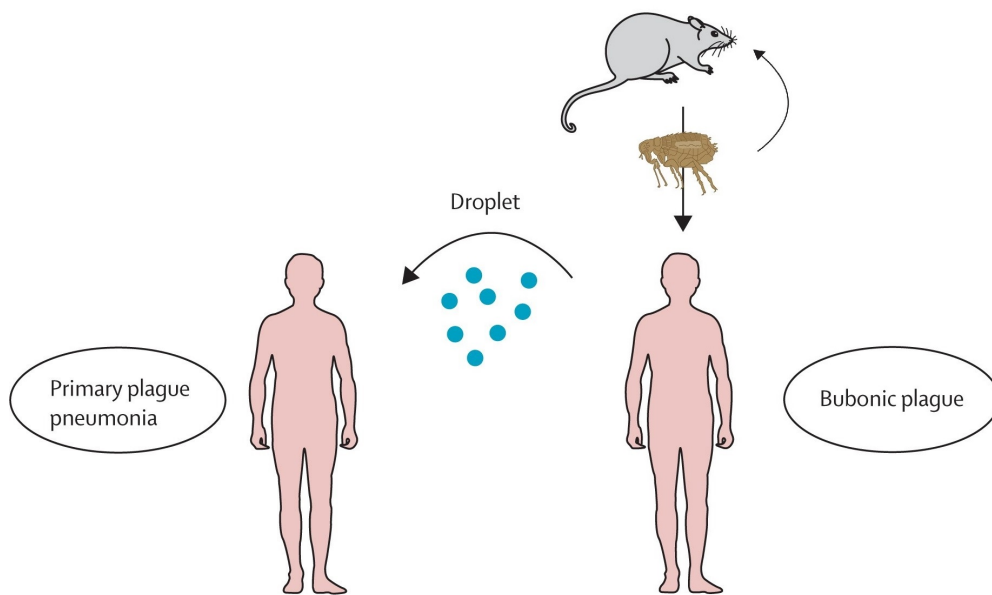


acceleration: conclusions

- wills seem to be an adequate (if noisy) proxy for parish records/plague-specific mortality records (London Bills)
- epidemic speed (r) increased from $\approx 10/\text{year}$ to $20/\text{year}$ (doubling time from ≈ 4 weeks to ≈ 2 weeks)

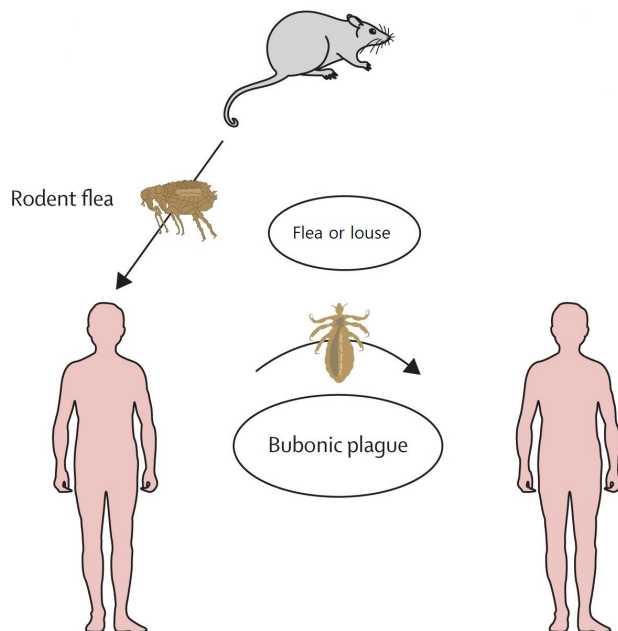
Y. pestis transmission

- Classic story: rat-flea transmission, human spillover
- Possible *pneumonic* transmission

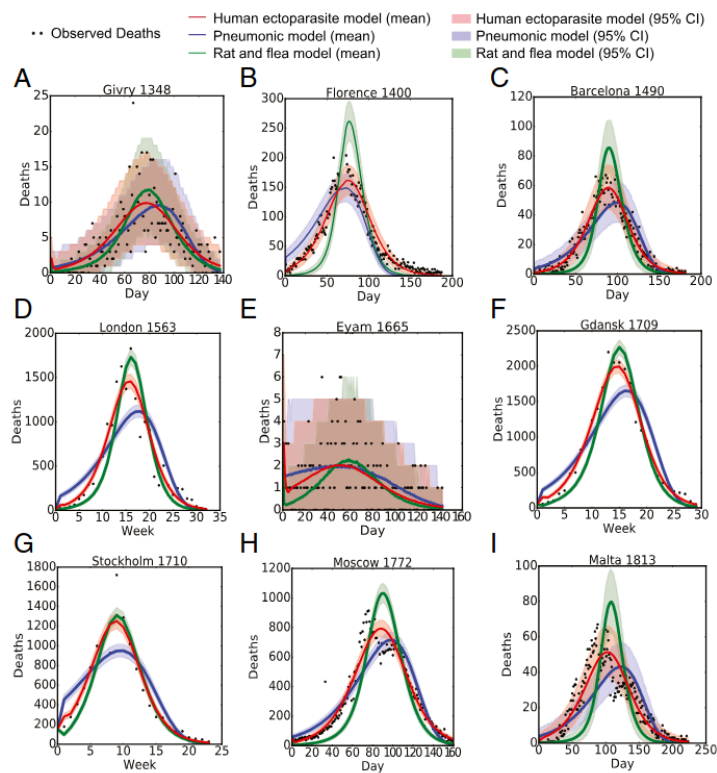


Human-ectoparasite transmission

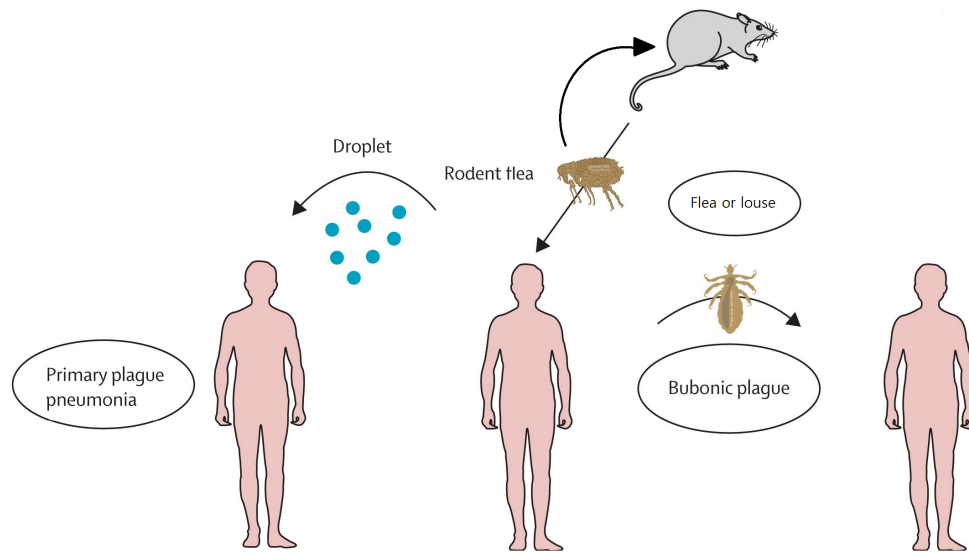
Human-to-human transmission via ectoparasites (e.g. lice)



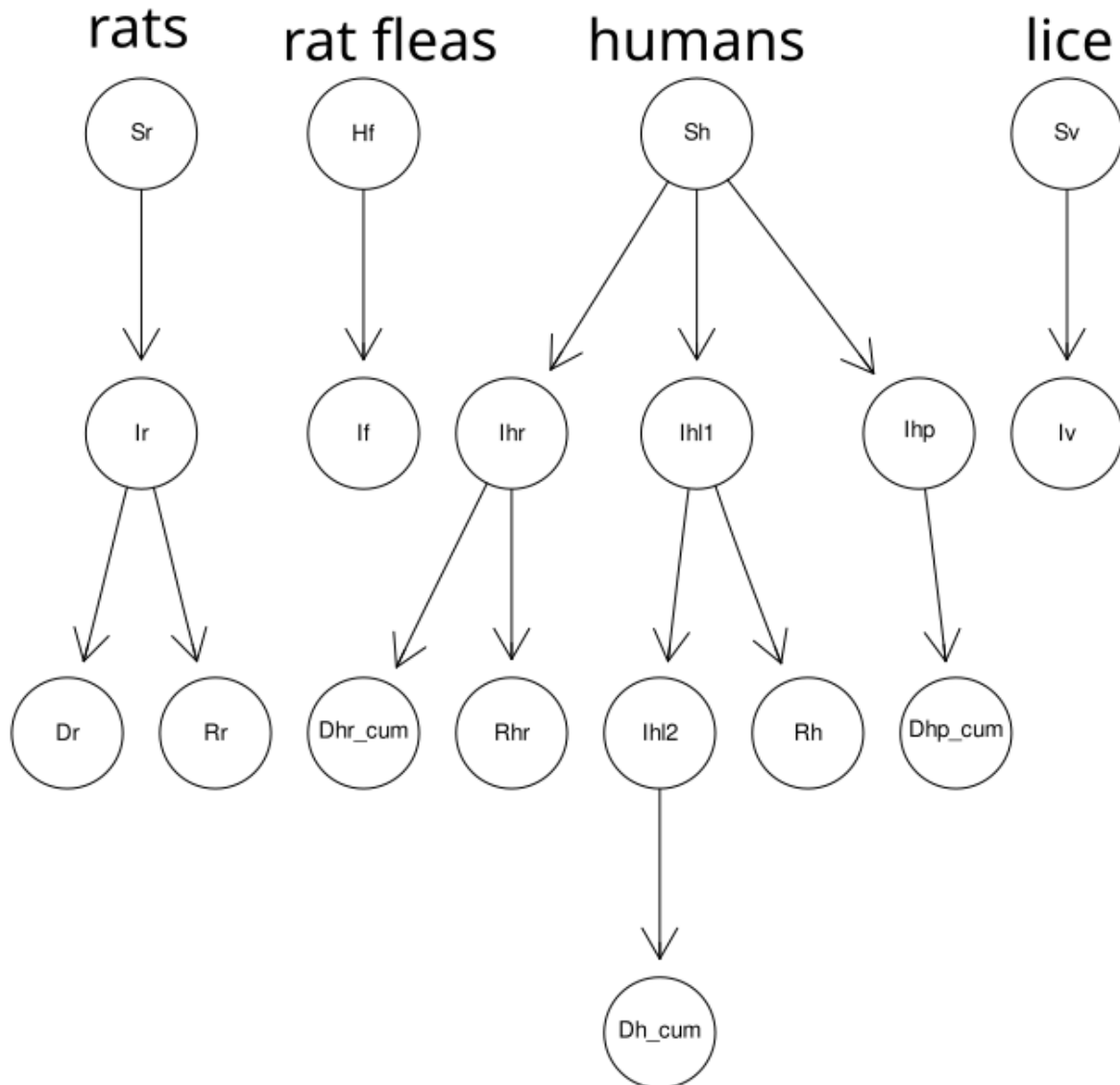
data fits (Dean et al., 2018; Dean, 2019)



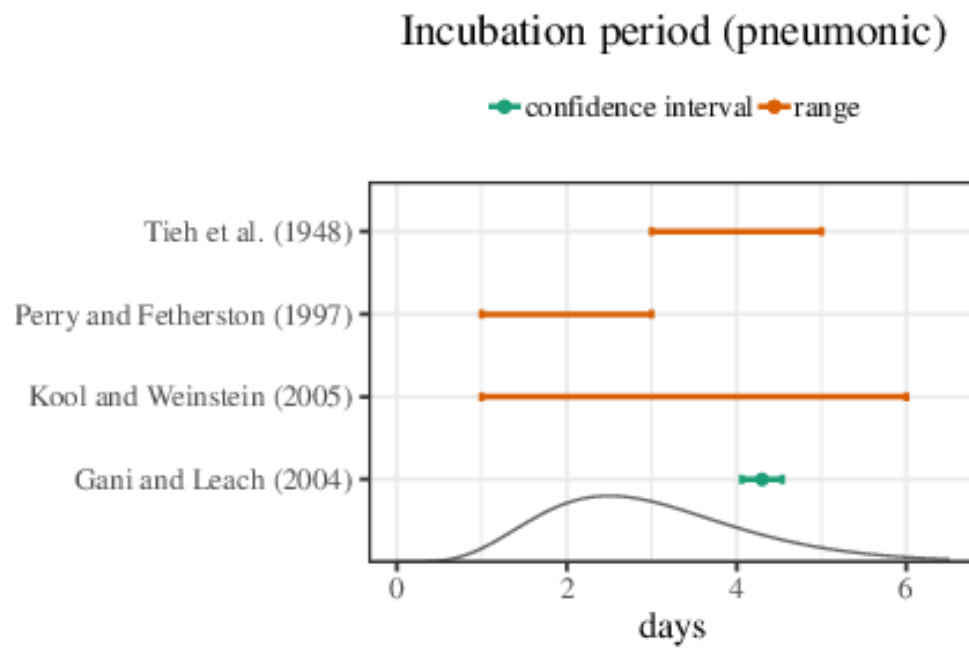
combined model



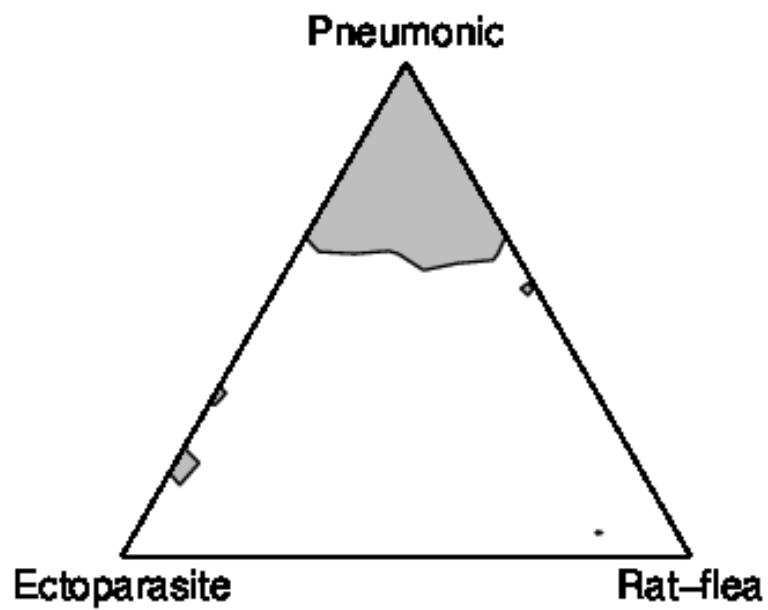
box diagram



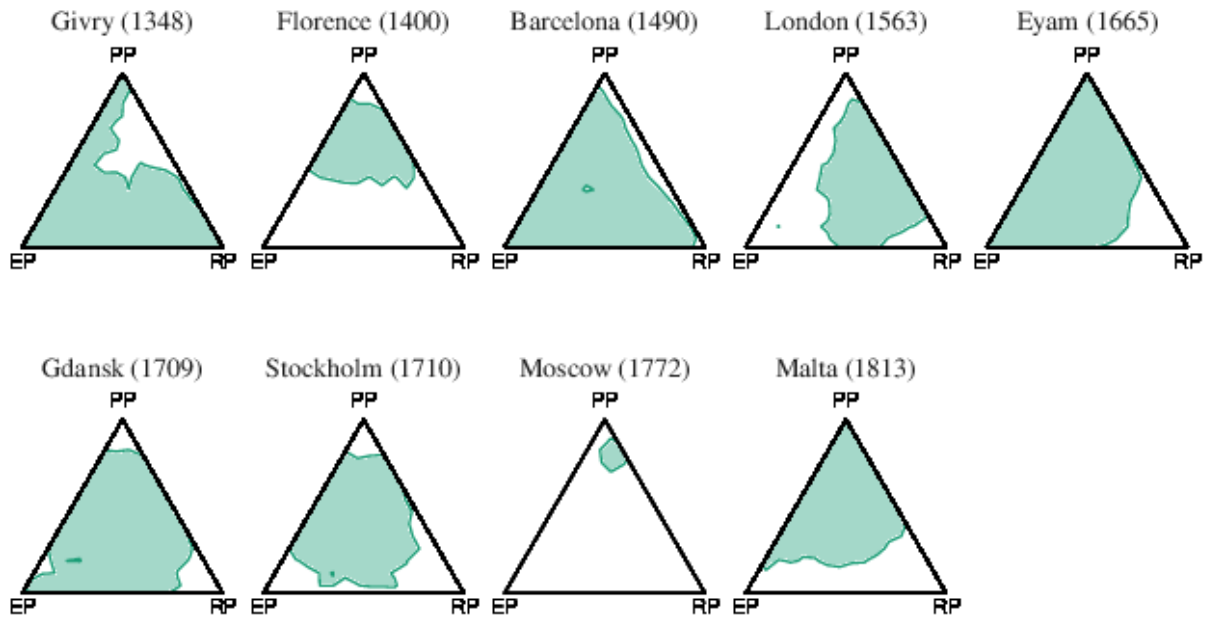
priors



results (example)



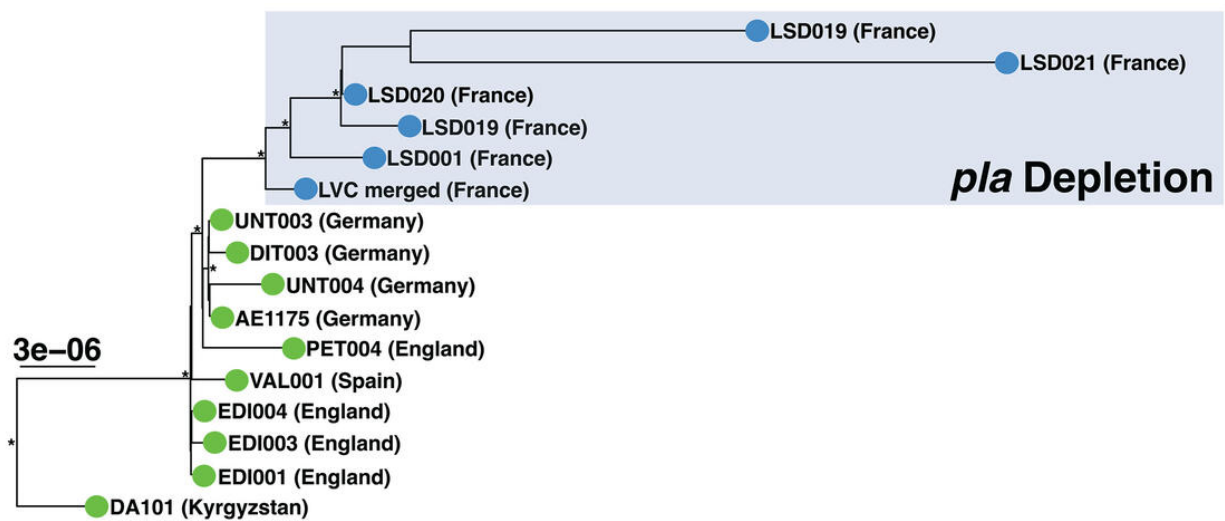
results (all)



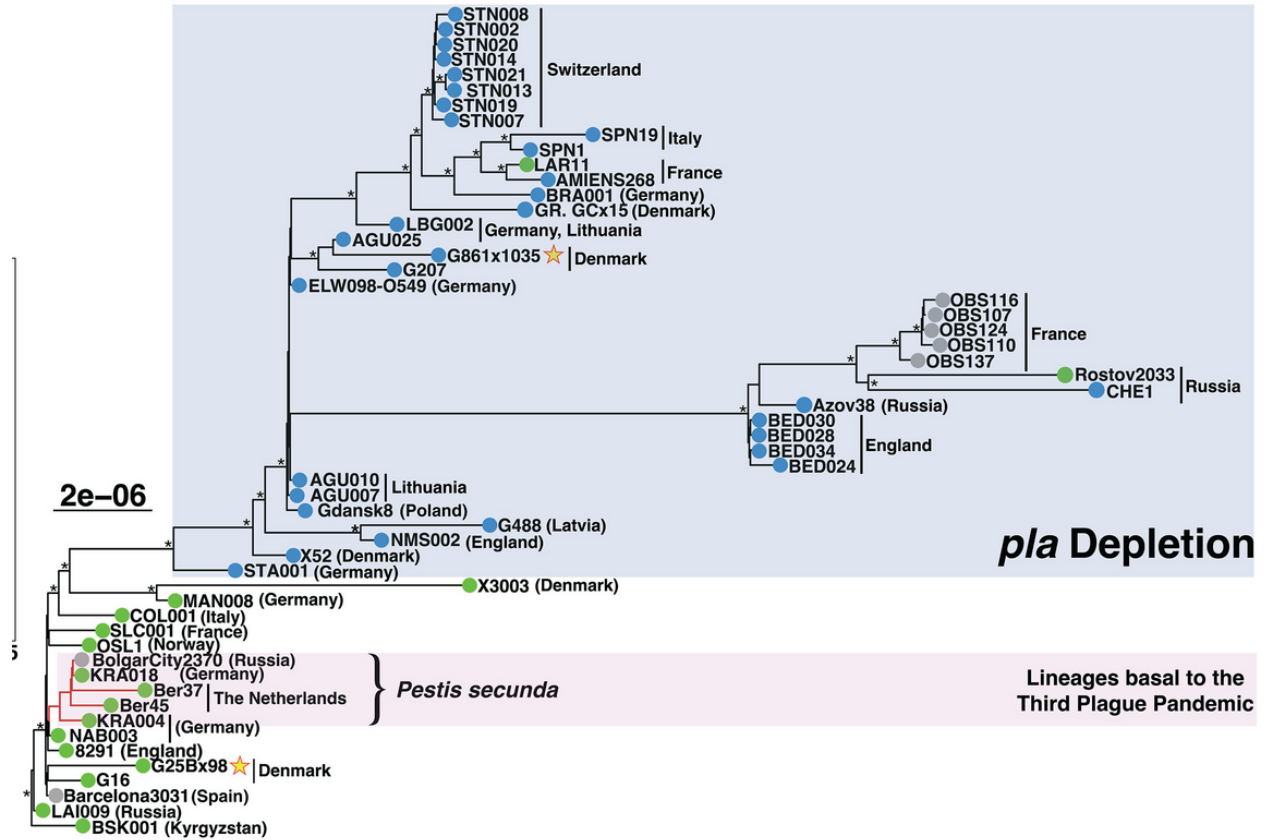
virulence evolution

- Sidhu et al. (2025)
- repeated evolution of PLA (plasminogen activator) depletion across pandemics

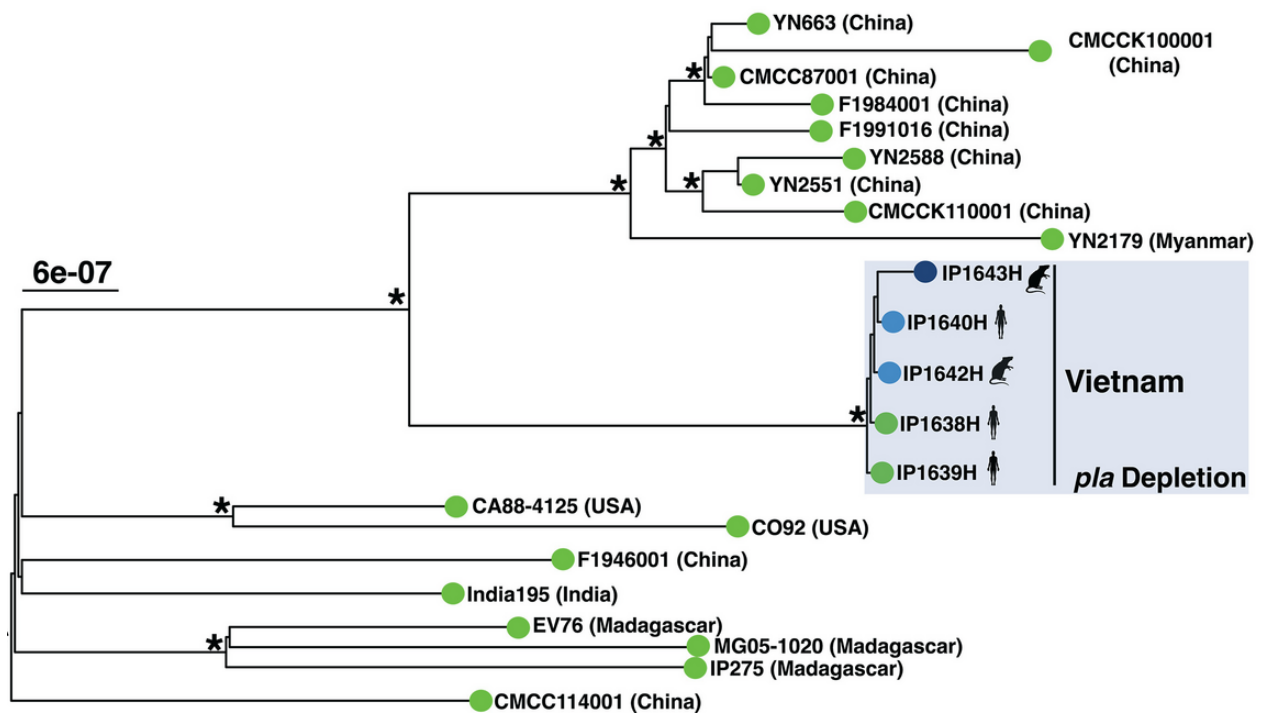
PLA results: first pandemic



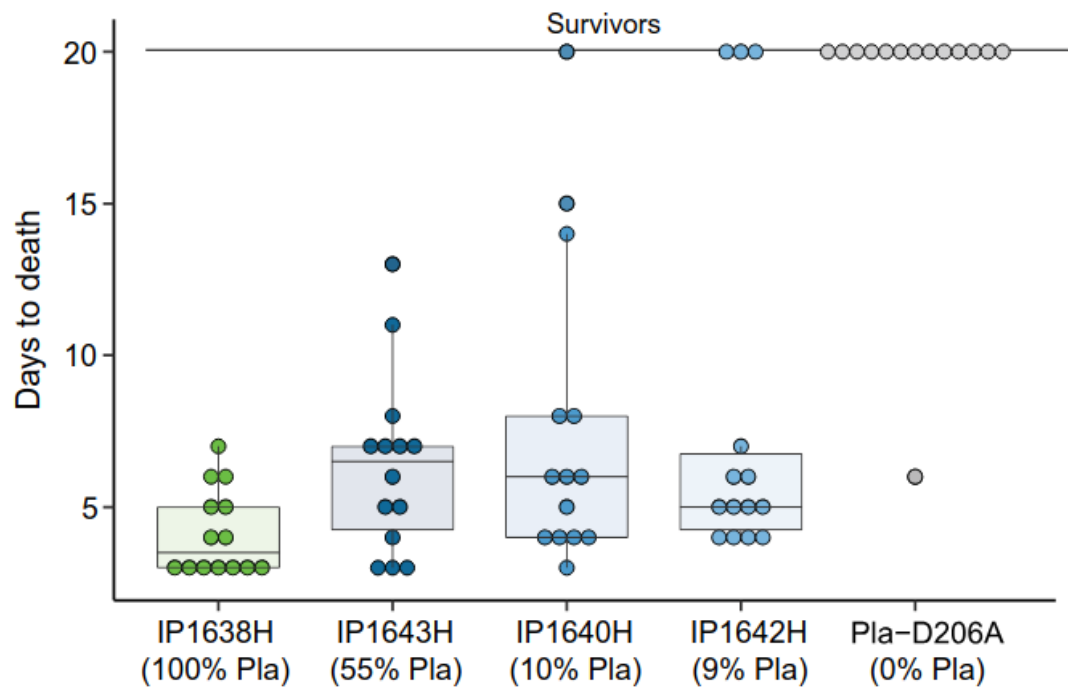
second pandemic



third pandemic



PLA phenotypes



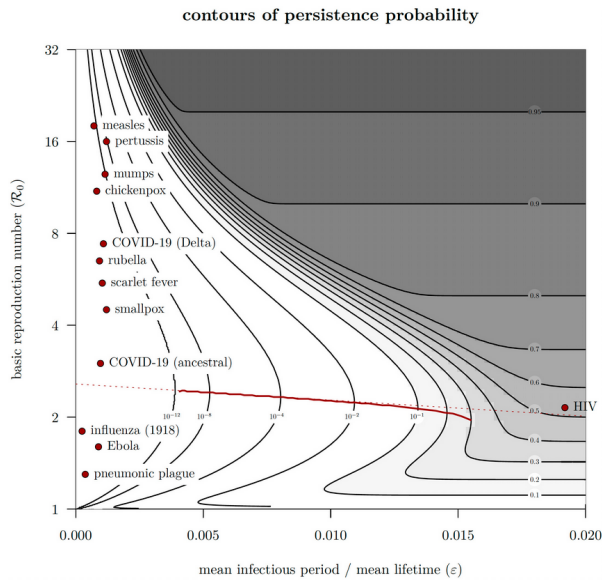
evolutionary mechanisms?

- tradeoff theory *doesn't work*

- most transmission is via fleas leaving dying rats
- ‘infectious period’ is a burst at the end of the infection
- how can PLA-depleted strains evolve (repeatedly)?

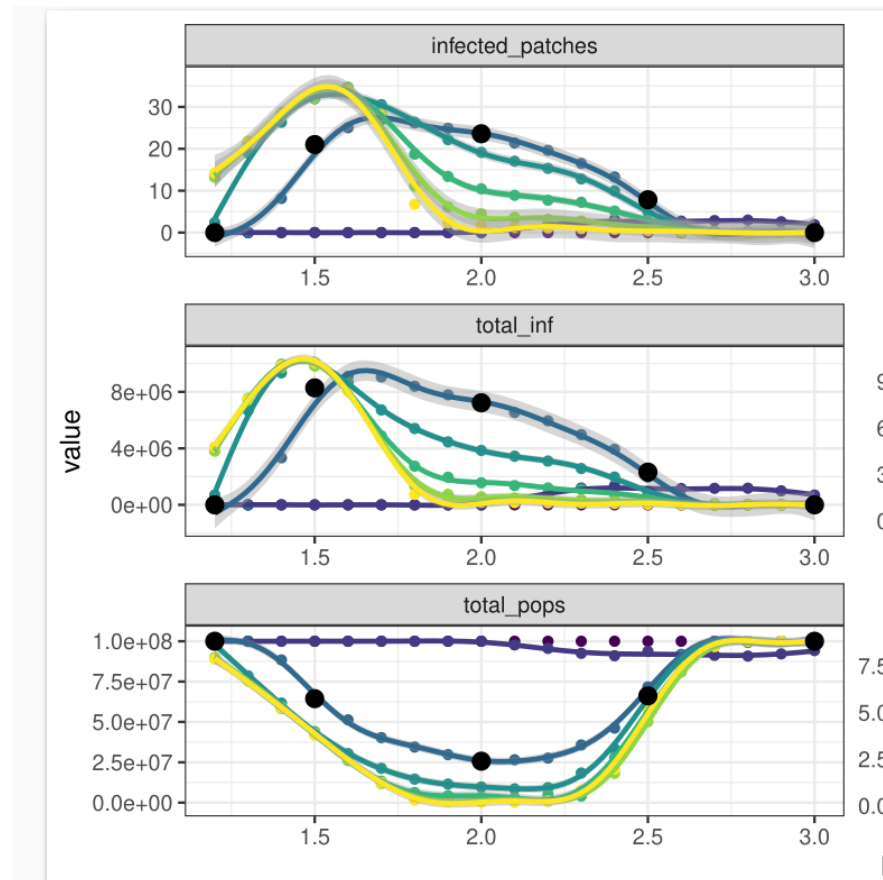
burnout theory

Parsons et al. (2024)



metapopulation model

metapopulation results



References

- Dean, K. R. (2019). *The epidemiology of plague in Europe: Inferring transmission dynamics from historical data* [Doctoral Thesis].
- Dean, K. R., Krauer, F., Walløe, L., Lingjærde, O. C., Bramanti, B., Stenseth, N. C., & Schmid, B. V. (2018). Human ectoparasites and the spread of plague in Europe during the Second Pandemic. *Proceedings of the National Academy of Sciences*, 201715640. <https://doi.org/10.1073/pnas.1715640115>
- Earn, D. J. D., Ma, J., Poinar, H., Dushoff, J., & Bolker, B. M. (2020). Acceleration of plague outbreaks in the second pandemic. *Proceedings of the National Academy of Sciences*, 117(44), 27703–27711. <https://doi.org/10.1073/pnas.2004904117>
- Harbeck, M., Seifert, L., Hänsch, S., Wagner, D. M., Birdsell, D., Parise, K. L., Wiechmann, I., Grupe, G., Thomas, A., Keim, P., Zöller, L., Bramanti, B., Riehm, J. M., & Scholz, H. C. (2013). *Yersinia pestis* DNA from Skeletal Remains from the 6th Century AD Reveals Insights into Justinianic Plague. *PLOS Pathogens*, 9(5), e1003349. <https://doi.org/10.1371/journal.ppat.1003349>
- Parsons, T. L., Bolker, B. M., Dushoff, J., & Earn, D. J. D. (2024). The probability of epidemic burnout in the stochastic SIR model with vital dynamics. *Proceedings of the National Academy of Sciences*, 121(5), e2313708120. <https://doi.org/10.1073/pnas.2313708120>
- Sidhu, R. K., Mas Fiol, G., Lê-Bury, P., Demeure, C. E., Bougit, E., Beau, R., Balière, C., Kwasiborski, A., Caro, V., Klunk, J., Salkeld, D. J., Carmichael, A., Varlik, N., Poinar, D., Earn, D. J. D., Bolker, B. M., Dushoff, J., Golding, G. B., Rascovan, N., ... Poinar, H. N. (2025). Attenuation of virulence in *Yersinia pestis* across three plague pandemics. *Science*, 388(6750), eadt3880. <https://doi.org/10.1126/science.adt3880>

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